UIU RESCUE ROVER

Complete Website Redesign Blueprint

From first loading frame to final footer: story, interaction, visual system, asset recovery, claims, engineering content, research evidence, accessibility and launch.

Version 1.0 | 26 October 2026 | Implementation specification / creative proposal

The experience in one sentence

A visitor enters a real rescue-robotics mission, discovers the people and engineering behind the machines, explores authentic prototypes, sees measurable evidence, and leaves with a clear next action.

Non-negotiable outcome

• No generic AI landing page. No fabricated robot photos, team logos, performance numbers, awards, publications or disaster deployments.

• Recover real media and official branding from the existing site before rebuilding hero and project pages.

• A cinematic, interactive homepage is the front door; research and technical pages remain readable, precise and source-linked.

• All claims must be categorized as verified, awaiting review, planned, or unsupported before publication.

Basis: user-provided DESIGN.md (Field Dossier, Archivo / Source Serif, paper + charcoal + lime, plate captions, keyboard access, one authored motion moment). This blueprint proposes changes where the current implementation lacked authentic assets and visual impact.

01 / The first 5 seconds: loading and entry

0–0.5 s — immediate first paint

Show the real official logo/wordmark, navigation shell, meaningful background and hero headline immediately. No blocking splash screen, fake percentage counter, terminal boot sequence or forced intro. Preload only critical fonts and the hero poster image. Preserve a stable layout to avoid visual jumping.

0.5–2 s — cinematic hero resolves

A real AEGIS or Zugbee X1 photograph fills 55–70% of the hero area (or an approved team video with a static poster). The hero copy is already readable. A subtle fade of image detail is allowed; never hide primary content behind animation. If no suitable authentic image is available, use a typographic editorial hero with an honest, labelled technical figure rather than an invented rover.

First screen — precise content

• Brand: actual UIU Rescue Rover logo; university attribution secondary.

• Headline proposal: “When humans cannot enter, robots go first.” This is aspirational positioning, not a claim of operational deployment.

• Supporting line: “Designing and testing robotic systems for challenging search-and-rescue environments.”

• Primary CTA: Explore our robots. Secondary CTA: See our experiments.

• Quiet cue: Scroll to explore. Keep navigation visible but unobtrusive.

Performance constraints

Use an optimized responsive poster, local media where permitted, optional muted/looping video only after interaction or when bandwidth allows. Respect reduced-motion and save-data settings. Never autoplay audio. Test on midrange Android and slow mobile networks.

02 / Homepage: scene-by-scene storyboard

01. THE MISSION

Dark cinematic opening; authentic rover close-up; one bold headline; two calls to action. Interaction: subtle reveal, no scroll hijacking.

02. THE HUMAN PROBLEM

Editorial sequence showing real rescue challenges: rubble, access, visibility, communications. Use sourced imagery/diagrams, not fabricated real missions. Interaction: tap challenges to reveal engineering implications.

03. THE FIRST PROTOTYPE

Zugbee X1 origin: what the team wanted to learn, what was built, what changed. Real photos + caption + dated milestones. Interaction: scrub verified build stages.

04. THE EVOLUTION

Transition to AEGIS only where chronology is documented. Before/after component comparison, design trade-offs and changes. Interaction: click annotated hotspots.

05. INSIDE THE MACHINE

Mechanical, power, sensing, communication, control and autonomy layers. Interactive annotated photo/SVG with accessible keyboard tabs. Every component has actual status and evidence.

06. BUILT / TESTED / MEASURED

Show selected real experiments: methodology, conditions, trials, results and failures. Filter by subsystem and robot. Link to full records and data.

07. THE PEOPLE

Human-centered photography and honest team contributions; advisor attribution and verified roles. Interaction: filter by discipline, not gimmicky 3D flipping cards.

08. THE JOURNEY

Chronological verified events, competition participation, build milestones, failures and future goals. Never auto-convert past dates into success claims.

09. JOIN THE WORK

Collaboration, recruitment, sponsors, open resources. One concise primary CTA and clear contacts; no fake sponsor logos.

10. THE FOOTER

Official branding, direct page links, social links only when verified, contact, copyright, accessibility, last-updated and content corrections link.

03 / Visual and interaction direction

Visual grammar

• Hero and evidence climax: charcoal cinematic surfaces. Research and long-form content: warm paper surfaces. Avoid alternating every section mechanically.

• Palette starting point: #101820 charcoal, #F5F4EF warm paper, #B7F36B lime accent, #46525C secondary ink. Validate contrast.

• Typography: Archivo for display/UI; Source Serif 4 for editorial body, per supplied DESIGN.md. Use the serif selectively; avoid dense paragraphs in the hero.

• Large authentic photography with disciplined crops and factual captions. Editorial grids, asymmetry, restrained rules, no endless identical cards.

• Use real team logo and official visual assets. Do not redraw an unofficial logo as if official.

Purposeful interactions

• Prototype timeline: accessible slider/step controls + scroll-linked visual progression; static chronological fallback.

• Robot explorer: annotated real image or SVG with click, keyboard, touch; details panel with component role, specs, limitations and source.

• Experiment explorer: searchable filters, shareable URL state, comparison charts only with real data, download links when permitted.

• Engineering story: before/after changes, design trade-off tabs, failure-analysis reveals; each interaction adds information.

• Motion: mostly 120–420 ms transitions, transform/opacity, no infinite animations or forced scrolling.

What to avoid

No stock photos masquerading as UIU robots; no generic tracked chassis in the hero; no invented live telemetry; no unverified numerical counters; no public “10/11 claims under review” dashboard; no heavy WebGL just to appear modern.

04 / Asset recovery and migration: mandatory first sprint

Sources to inspect

Start with https://rescuerover.uiu.ac.bd/index.html and crawl discoverable internal pages, including project, member, advisor, contact, event and sponsor pages. Use MarkCrawl for content inventory, plus HTML inspection / browser rendering for lazy-loaded images, CSS backgrounds, srcset, video embeds, SVGs and favicons.

Required asset manifest

Field	Meaning
asset_id	Stable identifier
original_url	Exact URL on the source site
source_page	Page where discovered
subject	AEGIS / Zugbee X1 / member / event / logo
license_or_permission	Reuse authorization / pending review
local_path	Production asset destination
caption_and_alt	Factual caption and accessible description
status	verified / review / rejected

Priority retrieval

• P0: official logo, favicon, actual rover hero photograph and project-gallery images.

• P1: team and advisor portraits, workshop/testing footage, competition photos and original captions.

• P2: sponsor marks, publications, diagrams, technical documentation, approved video embeds.

Validate actual HTTP response, MIME type, dimensions, image decoding and identity. Deduplicate by file hash; preserve originals where allowed and create responsive derivatives. If media cannot be recovered, document the failure and request the files from the team. Never silently substitute unrelated imagery.

05 / Information architecture and page contracts

Route	Must contain
/	Complete narrative and clear next actions
/robots	Prototype comparison, evidence-backed specifications
/robots/[slug]	Gallery, architecture, version, constraints, changelog
/research	Research questions, methods, publications/status
/research/[slug]	Research overview, linked experiments and citations
/experiments	Filterable experiment index with real data
/experiments/[slug]	Protocol, setup, trials, metrics, failures, assets
/engineering	Subsystem breakdown and design decisions
/journey	Verified timeline, participation vs results
/people	Team roles, contributions and advisors
/resources	Publications, code, data, media and reuse terms
/join /collaborate /contact	Actionable forms and verified contact details

Relationships

Robot → subsystem → design decision → experiment → result → evidence file; member → verified role → contribution; competition → participation → outcome. Reuse a structured content model rather than duplicating facts across pages.

06 / Claims: public trust and internal review

Claim policy

• VERIFIED: source evidence checked by a responsible team owner. Publicly display with appropriate citation/date.

• NEEDS REVIEW: do not display as a confident factual statement; request verification internally.

• PLANNED: label explicitly as goal, prototype intention or future work, never achieved capability.

• UNSUPPORTED: remove from publication until evidence exists.

High-risk claims to audit

Claim type	Required evidence / treatment
Robot autonomy	Demonstration conditions, operator involvement, repeatable tests
Victim detection	Dataset, protocol, false positives/negatives, limitations
Rescue readiness	Deployment certification or field validation; otherwise avoid
Success rate / test hours	Raw logs, definition, trial count, dates
Awards / rankings	Official organizer results and correct team attribution
Publications	DOI, venue, status; submitted ≠ accepted
Team size / members	Current roster owner, last verification
Partners / sponsors	Permission to use marks and correct relationship

Example content record

Claim ID: aegis-mobility-001 | Text: “AEGIS traversed obstacle type X under condition Y” | Evidence: test video + dated protocol + CSV | Trials: N | Owner: mechanical lead | Reviewed: YYYY-MM-DD | Status: verified. Never fabricate X, Y or N.

Public vs internal

The public website should show source-backed facts and limitations. An internal admin view can show claim counts, owners and outstanding reviews. Do not place editorial backlog numbers in the homepage hero or footer.

07 / Research credibility: experiment detail template

Research question

Precise falsifiable question, why it matters, related robot/subsystem.

Method and setup

Hardware revision, firmware/software versions, calibration, terrain, environmental conditions, operator role.

Protocol

Steps, inclusion criteria, stopping rule, baseline, number of trials, success definition.

Results

Raw measurements, summary statistics, uncertainty, plots and relevant failures; no invented values.

Evidence

Original video, logs, code, dataset, photos, schematic and downloadable report where approved.

Interpretation

What results support; what they do not support; confounders and generalization limits.

Reproducibility

Configuration files, dataset schema, dates, contributor and revision history.

Dashboard guardrails

A chart renders only when verified numeric data exists. Empty datasets show “No validated measurements published” with an explanation, not a decorative fake graph. Label simulation, replay and real robot telemetry distinctly.

08 / Implementation, QA and acceptance criteria

Recommended stack

Next.js + TypeScript, Tailwind or consistent design tokens, MDX for initially developer-maintained content, versioned JSON/CSV for experiments, Motion for limited transitions, optional React Three Fiber only after a genuine 3D asset and clear interaction need. Use GitHub Actions for builds, lint, links and content validation.

Build sequence

- Sprint 0: crawl old site, recover official logo/media, make asset and claim manifests. No final hero before this.
- Sprint 1: visual direction, storyboards, mobile/desktop prototypes, team approval.
- Sprint 2: homepage and core navigation with authentic media, real responsive behavior.
- Sprint 3: robot pages, prototype timeline, annotated robot explorer.
- Sprint 4: research/experiment schemas, verified records, resource pages, people and journey.
- Sprint 5: content review, browser QA, accessibility, performance, SEO, redirects and launch.

Launch gates

- Every visible photograph correctly identified and permitted; logo is authentic.
- All major claims checked or explicitly excluded; no fabricated statistics or achievements.
- All navigation and CTAs work; no dead buttons; no broken images.
- Mobile, tablet and desktop tested; keyboard/touch and reduced-motion work.
- Production build succeeds; basic security and form validation checked.
- Lighthouse and real-device performance measured; image and video payload reviewed.
- Existing URL redirects and rollback plan prepared; ownership for ongoing updates assigned.

Ownership and maintenance

Project lead approves project status; research lead reviews technical claims; team coordinator maintains people; competition coordinator maintains events; site maintainer manages deployment. Quarterly content audit, and immediate updates after verified major milestones.

09 / Final creative acceptance checklist

FIRST FRAME

Is the team unmistakably recognizable? Is the hero a real asset, or an honestly labelled alternative?

STORY

Can a new visitor explain the mission, prototype evolution, test evidence and team after scrolling?

INTERACTION

Does every animation or click teach something, and does it work on touch and keyboard?

RESEARCH

Can a skeptical researcher trace major claims to methods, results and sources?

TRUST

Are future goals separated from completed work and other teams’ achievements?

DESIGN

Is the page memorable without relying on stock photos, fake metrics, gradients or generic cards?

OPERATIONS

Can team members keep the website current without a developer rewriting every page?

The exact deliverables to request from the coding agent

- Source-site crawl report and asset manifest with verified image/logo files.
- Desktop and mobile wireframes for all ten homepage scenes.
- Working Next.js implementation with the real media integrated.
- Robot explorer, prototype timeline and experiment explorer with accessible controls.
- Content claim ledger, structured experiment schema and editorial review process.
- Screenshots, QA report, production build results, deployment/rollback instructions.

This is a design and implementation specification, not a claim that original media has already been downloaded or that any particular robot performance has been verified.